

Large-Scale Movements of IF3 and tRNA during Bacterial Translation Initiation

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In bacterial translational initiation, three initiation factors (IFs 1-3) enable the selection of initiator tRNA and the start codon in the P site of the 30S ribosomal subunit. Here, we report 11 single-particle cryo-electron microscopy (cryoEM) reconstructions of the complex of bacterial 30S subunit with initiator tRNA, mRNA, and IFs 1-3, representing different steps along the initiation pathway. IF1 provides key anchoring points for IF2 and IF3, thereby enhancing their activities. IF2 positions a domain in an extended conformation appropriate for capturing the formylmethionyl moiety charged on tRNA. IF3 and tRNA undergo large conformational changes to facilitate the accommodation of the formylmethionyl-tRNA (fMet-tRNA(fMet)) into the P site for start codon recognition.